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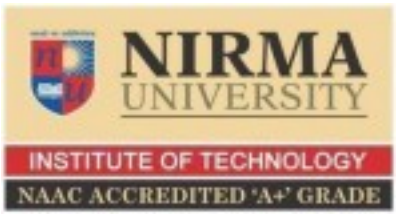
Software Project Management - 4CS504ME25 B.

Tech. Semester IV

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September 2026

Aim

To estimate software effort using the Use Case Point (UCP) method and perform a quantitative Make-vs-Buy decision analysis for the Hospital Management System (HMS).

Objective

- Classify system actors into Simple, Average and Complex categories to compute Unadjusted Actor Weight (UAW).
- Categorize use cases by transaction complexity to determine Unadjusted Use Case Weight (UUCW) and total UUCP.
- Score the 13 Technical Complexity Factors (TCF) and 8 Environmental Factors (EF).
- Calculate final adjusted Use Case Points and convert them into person-hours and person-months of effort.
- Isolate the Payment Gateway and Patient Notification Subsystem for Make-vs-Buy evaluation.
- Compare 5-year Total Cost of Ownership (TCO) between in-house build and commercial SaaS licensing.
- Find the break-even year and score options using a Weighted Decision Matrix.

1. Introduction

This report extends our earlier Hospital Management System study by applying two techniques used in software project management: Use Case Point (UCP) estimation for sizing and effort and Make-vs-Buy analysis for choosing how a system component should be delivered.

UCP was created by Gustav Karner to size object-oriented systems directly from use cases and actors. Because it looks at user interactions rather than lines of code, it works early in the project lifecycle once requirements and use case models are documented.

Make-vs-Buy analysis helps managers decide whether to build a feature with the internal team or license an existing cloud service. While our core clinical logic is custom built, integrating external payment processing and SMS alerts presents a realistic trade-off between build cost, speed and maintenance burden.

2. Use Case Point (UCP) Estimation

The UCP method calculates software size and effort through five steps based on system actors, use cases and project conditions.

2.1 Step 1: Identify and Classify Actors

Actors interact with the system and are classified by interface type:

- Simple (Weight 1, Count 2): Third-party Payment Gateway API and SMS Notification API.
- Average (Weight 2, Count 1): Pharmacy Barcode Scanner hardware interface.
- Complex (Weight 3, Count 4): Patient, Doctor, Receptionist/Staff and Hospital Administrator.

Actor Type	Definition / Interaction Method	Weight	Count	Total Weight
Simple	Interacts via programmatic API or background service	1	2	2
Average	Interacts via protocol or device (barcode scanner)	2	1	2
Complex	Human user interacting via web graphical UI	3	4	12

Actor Type	Definition / Interaction Method	Weight	Count	Total Weight
Total UAW	**Unadjusted Actor Weight (UAW) = 2 + 2 + 12**	**-**	**7**	**16**

2.2 Step 2: Identify and Classify Use Cases

Use cases are classified based on the number of transactions and steps involved:

- Simple (Weight 5, Count 3): User Login and Profile, View Doctor Slot Availability, View Prescription History.
- Average (Weight 10, Count 5): Register New Patient, Manage Doctor Roster, Book Appointment Slot, Manage Medicine Inventory, Generate Billing Invoice.
- Complex (Weight 15, Count 3): Inpatient Admission and Discharge, Complete Checkout and Payment Settlement, Hospital KPI Analytics.

Use Case Category	Transaction Complexity	Weight	Count	Total Weight
Simple	<= 3 transactions (auth, lookup, view history)	5	3	15
Average	4 to 7 transactions (register, book slot, inventory)	10	5	50
Complex	> 7 transactions (admission, checkout, analytics)	15	3	45
Total UUCW	**Unadjusted Use Case Weight (UUCW) = 15 + 50 + 45**	**-**	**11**	**110**

Unadjusted Use Case Points: UUCP = UAW + UUCW = 16 + 110 = 126.0 UUCP

2.3 Step 3: Technical Complexity Factor (TCF)

Thirteen technical factors evaluate system architecture, response time, reusability and security:

Factor	Description	Weight	Rating (0-5)	Weighted Value
T1	Distributed system (Express backend and React UI)	2.0	3	6.0
T2	Response time and performance objectives	1.0	3	3.0
T3	End-user efficiency (fast receptionist booking)	1.0	4	4.0
T4	Complex internal processing (billing logic)	1.0	3	3.0
T5	Reusability of code components	1.0	3	3.0
T6	Ease of installation and local setup	0.5	3	1.5
T7	Ease of use and UI ergonomics	0.5	4	2.0
T8	Portability across browsers and OS	2.0	3	6.0
T9	Ease of change and maintenance	1.0	3	3.0
T10	Concurrency (simultaneous receptionist and doctor access)	1.0	3	3.0

Factor	Description	Weight	Rating (0-5)	Weighted Value
T11	Special security features (medical records auth)	1.0	3	3.0
T12	Direct access for third parties (payment webhook)	1.0	2	2.0
T13	Special user training facilities	1.0	1	1.0
Total	**Sum of weighted technical factors: sum(TFactor)**	**-**	**-**	**40.5**

TCF = 0.60 + (0.01 * sum(TFactor)) = 0.60 + (0.01 * 40.5) = 1.005

2.4 Step 4: Environmental Factor (EF)

Eight environmental factors assess team skill, motivation, stability and tools:

Factor	Description	Weight	Rating (0-5)	Weighted Value
E1	Familiarity with development process	1.5	3	4.5
E2	Application experience in web systems	0.5	3	1.5
E3	Object-oriented and component experience	1.0	4	4.0
E4	Lead analyst capability	0.5	4	2.0
E5	Team motivation and collaboration	1.0	4	4.0
E6	Stability of requirements	2.0	3	6.0
E7	Part-time staff (student schedule)	-1.0	2	-2.0
E8	Difficult programming language	-1.0	1	-1.0
Total	**Sum of weighted environmental factors: sum(EFactor)**	**-**	**-**	**19.0**

EF = 1.40 - (0.03 * sum(EFactor)) = 1.40 - (0.03 * 19.0) = 1.40 - 0.57 = 0.83

2.5 Step 5: Final UCP, Effort and Cost

Adjusted Use Case Points: UCP = UUCP * TCF * EF = 126.0 * 1.005 * 0.83 = 105.10 UCP

Using the standard 20 person-hours per UCP:

Total Effort = 105.10 UCP * 20 Hours/UCP = 2,102 Person-Hours

At 160 hours per month: Effort = 2,102 / 160 = 13.14 Person-Months

At our student project labor cost of ₹20,000 per person-month, total development cost is about ₹2,62,800. This 13.14 PM figure covers full commercial production rules, whereas the COCOMO estimate in T7 (8.42 PM) corresponds to our academic prototype scope.

3. Make-vs-Buy Analysis: Payment and Notification Subsystem

3.1 Context and Selected Subsystem

Core hospital modules (Patient records, Doctor schedules, Pharmacy inventory and Billing records) must be developed in-house to protect patient privacy. However, processing credit card payments and sending SMS

appointment reminders requires external banking gateways and telecom integrations, making it a genuine make-vs-buy decision.

3.2 Subsystem Effort Derived from UCP

The payment and notification logic corresponds to two simple actors (Payment API and SMS API) and two use cases (Complete Checkout and Payment Settlement, plus View Prescription History notifications). This represents 21 UUCP:

Subsystem UCP = 21 * 1.005 * 0.83 = 17.51 UCP

Subsystem Effort = 17.51 * 20 = 350 Person-Hours = 2.19 Person-Months

Engineering labor cost to build this module internally is about ₹43,800. In addition, banking compliance, encryption and webhook infrastructure add ₹76,200, bringing total initial build cost to ₹1,20,000.

3.3 Option A: Build In-House

- Initial Development: Coding banking protocol drivers, webhook verifications, encryption and PCI-DSS compliance audits costs ₹1,20,000.
- Recurring Maintenance: Security audits, banking protocol updates and patch maintenance cost ₹40,000 per year.

3.4 Option B: Buy or License SaaS (e.g., RazorpayX and Twilio Health)

- Initial Integration: Quick SDK integration and webhook routing costs ₹25,000.
- Recurring Fees: Scaled per transaction. As patient volume grows from 5,000 in Year 1 to 30,000 in Year 5, annual costs rise from ₹30,000 to ₹1,80,000.

3.5 Five-Year TCO and Break-Even Calculaton

Year	Build In-House (Annual)	Build (Cumulative)	Buy SaaS (Annual)	Buy (Cumulative)
Year 0 (Setup)	₹1,20,000 (Dev cost)	₹1,20,000	₹25,000 (API setup)	₹25,000
Year 1	₹40,000 (Maintenance)	₹1,60,000	₹30,000 (5k txns)	₹55,000
Year 2	₹40,000 (Maintenance)	₹2,00,000	₹48,000 (8k txns)	₹1,03,000
Year 3	₹40,000 (Maintenance)	₹2,40,000	₹78,000 (13k txns)	₹1,81,000
Year 4	₹40,000 (Maintenance)	₹2,80,000	₹1,20,000 (20k txns)	₹3,01,000
Year 5	₹40,000 (Maintenance)	₹3,20,000	₹1,80,000 (30k txns)	₹4,81,000

Break-Even Calculaton:

- In Years 1 to 3, Buy is cheaper because of low initial cost (₹1.81 lakh for Buy compared to ₹2.40 lakh for Build at the end of Year 3).
- In Year 4, the cumulative cost of Buy (₹3.01 lakh) crosses Build (₹2.80 lakh).

Cost gap at end of Year 3 = ₹2,40,000 - ₹1,81,000 = ₹59,000 in favor of Buy.

Year 4 annual cost difference = ₹1,20,000 - ₹40,000 = ₹80,000.

Break-even Point = Year 3.0 + (₹59,000 / ₹80,000) = Year 3.0 + 0.74 = Year 3.74 (about 3 years and 9 months).

3.6 Qualitative Comparison and Decision Matrix

Regulatory compliance, security, maintenance and time to market must be weighed together:

Evaluation Criterion	Weight	Build Score (1-5)	Build Weighted	Buy Score (1-5)	Buy Weighted
Security and Regulatory Compliance (PCI-DSS and RBI)	30%	3	0.90	5	1.50
Time to Market and Integration Speed	20%	2	0.40	5	1.00
Maintenance and Support Burden	20%	2	0.40	4	0.80
Total Cost of Ownership (5-Year Horizon)	20%	4	0.80	3	0.60
Customization and Direct Control	10%	5	0.50	2	0.20
Total Weighted Score	**100%**	**-**	**3.00**	**-**	**4.10**

3.7 Recommendation

The Buy option scores 4.10 versus 3.00 for Build In-House. Banking and payment security require strict regulatory compliance that would introduce high operational risk if handled internally. Furthermore, Buy is financially cheaper for nearly four years. The practical decision is to license commercial payment and SMS APIs while keeping all core medical modules in-house.

4. Conclusion

The UCP method sized the Hospital Management System at 105.10 UCP (13.14 Person-Months), providing an objective estimate based on system interactions. Sizing the payment and notification module independently enabled a detailed Make-vs-Buy analysis. Combining the 5-year TCO comparison with a weighted decision matrix demonstrated that licensing commercial payment and notification services is the lower-risk choice, allowing our two-person team to focus on core hospital clinical workflows.